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Modal Analysis of Water Distribution Systems With the Elastic Water Column Model

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Key Points:

- The work reveals how resonance behavior in water distribution systems is spatially distributed, enabling identification of vulnerable areas
- This work provides a unified time-domain framework, contributing to improved system monitoring, management

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Abstract This paper presents a modal and participation factor (PF) analysis of water distribution systems (WDSs) using the elastic water column model (EWCm). Modal analysis, widely used in other engineering fields, is adapted here to characterize the dynamic behavior of WDSs under transient conditions. By linearizing the EWCm around an operating point, a state-space representation is developed, enabling the extraction of natural modes via eigenvalue analysis. These modes, defined by their frequencies and damping ratios, are validated through comparison with the admittance matrix method in the frequency domain. The study introduces PF analysis to quantify how each state variable (nodal head or flow rate) contributes to each mode. This spatial information identifies critical locations that are more sensitive to excitations and capable of amplifying transient responses. To verify the effectiveness of PF analysis, time-domain simulations are conducted for three test cases, including a real-world network (the New York tunnel system). The results confirm that exciting the system at high-PF locations can generate significant transients, while low-PF locations produce minimal responses. The analysis also reveals how resonance behavior in WDSs is spatially distributed, enabling the identification of vulnerable areas where transients are amplified. This work provides a unified time-domain framework for modal and PF analysis, contributing to improved system monitoring, management, and fault detection in WDSs.

Plain Language Summary Water distribution systems (WDSs) must be resilient to sudden changes like pump failures or shifts in demand, which can cause damaging pressure waves known as hydraulic transients. This study applies a method called modal analysis, commonly used in mechanical and electrical engineering, to better understand how WDSs behave under such conditions. Using a model that accounts for water compressibility—the Elastic Water Column Model—the researchers calculate the system's natural modes of vibration and how different parts of the system contribute to those vibrations. A key innovation is the use of participation factor (PF) analysis, which identifies the locations within the system most sensitive to dynamic disturbances. These critical points can amplify pressure surges if disturbed. By running simulations on both simple and complex networks, including the New York tunnel system, the study shows how PF analysis can predict where major transients might occur. This approach helps engineers identify weak points in large networks and improve system monitoring and control. Ultimately, it supports better design and operation of WDSs, making them more robust against disruptions. The methods used are efficient and can be integrated into existing modeling frameworks for real-world infrastructure planning and protection.

1. Introduction

Modal analysis is a widely used engineering technique for determining the dynamic characteristics of systems by identifying their natural frequencies, damping ratios, and mode shapes. It is particularly valuable in fields such as mechanical and civil engineering, where it helps predict how structures will respond to external forces, ensuring stability and identifying potential vulnerabilities. Modal analysis also plays a key role in the study of inter-area oscillations in power grids, helping to understand electro-mechanical oscillations (Dussaud, 2015; Klein et al., 1991). Furthermore, it has been extensively applied in aerospace, automotive design, and earthquake engineering, where understanding system dynamics is essential for both performance and safety (Dussaud, 2015; He & Fu, 2001).

Despite its broad use in other engineering disciplines, the application of modal analysis to WDSs remains limited. WDSs are subject to hydraulic transients caused by sudden changes in demand, pump failures, or valve operations. These transients can generate pressure surges that, if unmanaged, may damage infrastructure (Xing & Sela, 2020). Modal analysis offers a systematic framework for capturing the dynamic behavior of WDSs,

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extending beyond fault detection to improve the overall understanding of system response under various operating conditions.

Existing approaches for transient and modal analysis in WDSs can be broadly classified into time-domain simulation methods (e.g., the Method of Characteristics, MOC) and frequency-domain formulations (e.g., the transfer-matrix or admittance-matrix methods (AMM)) (Lee et al., 2005, 2006; Zecchin et al., 2009). Frequency-domain methods express the governing equations in the Laplace or Fourier domain to derive frequency response diagrams (FRDs), which relate the input and output spectra of pressure and flow signals. The FRD reveals resonance peaks that correspond to the natural frequencies of the system. The amplitude of each peak reflects the damping of its associated mode, with larger peaks indicating poles that lie closer to the imaginary axis and therefore represent more weakly damped oscillations (Zecchin et al., 2018).

In this regard, several studies have demonstrated that the FRD of a single pipeline consists of evenly spaced resonance peaks whose relative amplitudes depend on the measurement location and are sensitive to leaks or blockages. In particular, Lee et al. (2005), introduced inverse resonance and peak-sequencing methods to identify leaks by analyzing the pattern of resonant peaks in the FRD, while Louati et al. (2020) examined wave-leak interactions and showed that leaks modify the relative magnitudes of resonance peaks while leaving their frequencies largely unchanged. These studies established a clear frequency-domain understanding of pipeline modes and their role in resonance phenomena.

Although these methods have proven effective in controlled or single-pipe systems, they are fundamentally input-output based, relying on external excitation, transient simulation, or spectral analysis of measured signals to infer modal content. For example, one can use the MOC to generate pressure responses by applying a broadband excitation (e.g., a step or pulse) and then compute the corresponding frequency response function via Fast Fourier Transform (FFT) of the input-output signals (Lee Pedro et al., 2008). While this approach can reveal dominant resonance frequencies, it requires long-duration simulations, careful design of the input signal to excite all relevant modes, and sufficiently dense sensor coverage to capture spatial variations. Moreover, the quality of the inferred modes depends on the signal-to-noise ratio, sampling resolution, and how persistently each mode is excited. Repeating this process across many operating points or measurement pairs quickly becomes computationally demanding (Che et al., 2021).

In contrast, frequency-domain formulations such as the transfer-matrix method, determine resonance by directly solving the linearized equations in the frequency domain, but these methods provide only global input-output relationships and do not directly yield the internal modal structure, that is, the mode shapes or how each pipe or node contributes to each oscillation. Consequently, while FRD-based techniques offer valuable information on resonance frequencies, they lack the ability to describe the spatial distribution of modal energy within large, complex networks.

To overcome these challenges, this study introduces, for the first time, a state-space-based framework for modal analysis of WDSs using the Elastic Water Column Model (EWCm). The EWCm extends the Rigid Water Column Model (RWCM) by including the compressibility of water and pipe-wall elasticity through a set of ordinary differential equations (ODEs), enabling accurate simulation of fast transients while maintaining computational efficiency (Imani et al., 2025, 2026; Ulanicki & Beaujean, 2021; Zeng et al., 2022). Unlike input-output approaches such as the transfer-matrix method, the EWCm framework directly exposes the intrinsic dynamics of the linearized system. The eigenvalues of the system matrix correspond to the natural modes of the network, while the right and left eigenvectors describe how energy propagates through the network and how each state variable participates in each oscillation mode. This state-space formulation allows all modal information to be extracted in a single step, without requiring excitation signals, frequency sweeps, or curve fitting.

A key advantage of this formulation is its compatibility with Participation Factor (PF) analysis, which complements modal analysis by introducing spatial insight. PFs quantify the contribution of each state variable, nodal head or flow rate—to each mode and reveal which parts of the network are most dynamically involved. Originally developed for power-system stability studies (Abdulrahman, 2020; Abed et al., 2000), PF analysis bridges the gap between frequency and space: while frequency-domain diagrams show how a system behaves across frequencies, PFs show where each mode is most active within the network. This capability provides a powerful diagnostic tool for identifying spatially vulnerable regions that are prone to resonance or amplified transients, information that cannot be obtained from classical frequency-domain approaches.

This paper presents a time-domain approach to combined modal and PF analysis of WDSs using the EWCM. The integration of these techniques allows not only for the extraction of dynamic modes but also for the spatial identification of vulnerable locations where transients may be amplified. This enables a more complete understanding of how hydraulic transients affect WDSs and provides a new way to detect potential resonance behaviors in complex networks. The goals of this study are fourfold. First, a state-space representation of WDSs is developed by linearizing the EWCM around an operating point, enabling direct time-domain modal analysis. Second, the system's natural modes are extracted through eigenvalue analysis and validated against results obtained from the AMM in the frequency domain. Third, PF analysis is applied to quantify the spatial contributions of flow and head states to each mode, revealing critical locations that are more prone to amplifying transient responses. Finally, the practical implications of PF analysis are verified through time-domain simulations, including a real-world case study of the New York tunnel system subjected to broadband excitations.

After establishing and verifying the mathematical foundation of the proposed modal analysis framework, its potential applications are analogous to those of modal analysis in other engineering domains. By identifying dominant modes and their spatial characteristics, the framework can assist in surge protection assessment, guide sensor and actuator placement, and support the development of reduced-order models for monitoring and control. The New York tunnel test illustrates its practical use in a large-scale network, where complex interactions between topology and dynamics can obscure regions of vulnerability. Together, modal and PF analyses form a unified framework that links the mathematical structure of WDS dynamics to physically interpretable spatial behavior and provides a foundation for future applications such as real-time control and resilience assessment.

2. Elastic Water Column Model

The one-dimensional partial differential equations governing momentum and continuity in a pressurized pipeline system are expressed as:

$$\frac{\partial h}{\partial x} + \frac{1}{gA} \frac{\partial q}{\partial t} + \frac{f|q|}{2gDS^2} = 0 \quad (1)$$

$$\frac{gS}{a^2} \frac{\partial h}{\partial t} + \frac{\partial q}{\partial x} = 0 \quad (2)$$

where h represents the piezometric head, q is the volumetric flow rate, g is the gravitational acceleration, S is the internal cross-sectional area of the pipe, D is the pipe diameter, and f is the Darcy-Weisbach friction factor. The formulation of the EWCM originates from the analogy between hydraulic and electrical systems. The telegrapher's equation in electrical engineering shares the same mathematical form as the momentum and continuity equations in hydraulics. In this analogy, voltage corresponds to hydraulic head, current to flow rate, inductance to fluid inertia, resistance to pipe friction, and capacitance to the water compressibility. This correspondence forms the basis for representing hydraulic systems using an Electrical Equivalent Circuit (EEC) framework, which allows hydraulic energy storage, dissipation, and transfer to be analyzed in a physically intuitive way. When analyzing a pipe section of length l , under the assumption of constant hydraulic conditions (i.e., $\partial h/\partial x \approx \Delta h/l$ and $\partial q/\partial x \approx \Delta q/l$), the transient flow equations are reformulated following Souza et al. (1999):

$$\Delta h = -L \frac{\partial q}{\partial t} - Rq|q| \quad (3)$$

$$\Delta q = -C \frac{\partial h}{\partial t} \quad (4)$$

where L is the hydraulic inductance ($L = \frac{l}{gS}$), C is the hydraulic capacitance ($C = \frac{gS}{a^2}$), and R is the hydraulic resistance ($R = \frac{fl}{2gDS^2}$). Ideally, the flow rate change due to water compressibility should be distributed evenly along the pipeline, but the Electrical Equivalent Method (EEM) simplifies this by concentrating the changes at specific points within the pipe section. These points can be the center (T-shaped electrical circuit), the upstream end, downstream end, or both ends (π -shaped electrical circuit). Zeng et al. (2022) explored these configurations, and this study adopts the π -shaped electrical circuit for its efficiency in simulating the dynamic hydraulic

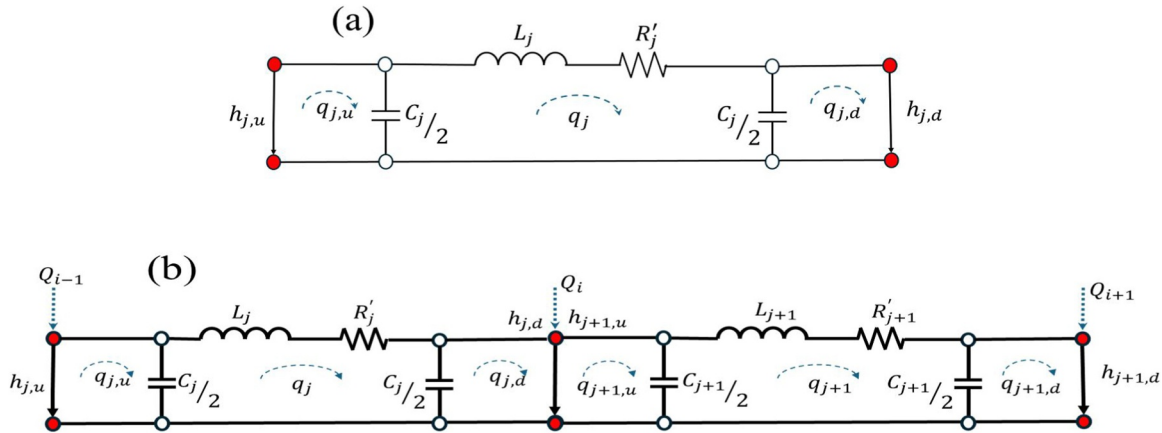


Figure 1. Electrical Equivalent Circuits: (a) EEC for a pipe segment; (b) EEC for two pipes in series.

behaviors with high accuracy. For such an EEM configuration, shown in Figure 1a, the governing equations for the j th pipe segment are:

$$R'_j = R_j |q_j| \quad (5)$$

$$h_{j,u} = L_j \frac{dq_j}{dt} + R'_j q_j + h_{j,d} \quad (6)$$

$$\frac{c_j}{2} \frac{dh_{j,u}}{dt} = q_{j,u} - q_j = -\Delta q_{j,u} \quad (7)$$

$$\frac{c_j}{2} \frac{dh_{j,d}}{dt} = q_j - q_{j,d} = -\Delta q_{j,d} \quad (8)$$

where the subscript j refers to the j th pipe segment in the network, and u and d represent the upstream and downstream ends, respectively. Equation 6 follows Kirchhoff's Voltage Law for the EEC in Figure 1a. In this equation, the sum of potential (head) losses due to inductance and resistance equals the head difference between two nodes. Using Equations 7 and 8, the continuity equation for a node, where multiple pipes converge, or demand exists, illustrated in Figure 1b is written as:

$$\sum_{j \in \Lambda_{i,d}} q_{j,d} - \sum_{j \in \Lambda_{i,u}} q_{j,u} + Q_i = 0 \quad (9)$$

where the sum $\sum_{j \in \Lambda_{i,d}} q_{j,d}$ represents the total inflow to node i from incoming pipes (set $\Lambda_{i,d}$), while $\sum_{j \in \Lambda_{i,u}} q_{j,u}$ represents the outflow from node i to outgoing pipes ($\Lambda_{i,u}$). Q_i represents the demand at node i . The sign of Q_i is positive where flow is directed toward the node, which is the opposite of the convention used by Zeng et al. (2022). This adjustment is made to maintain consistency with the AMM, where the flow toward nodes is considered positive to ensure that the system remains energy passive, that is, it does not generate energy, in a theoretically consistent sense (Zecchin, 2010). The terms $\Delta q_{j,u}$ and $\Delta q_{j,d}$ denote flow rate changes at the node due to water compressibility and pipe elasticity, affecting both upstream and downstream pipes. These changes are critical for understanding fluid dynamics within the network, especially in response to variations in pressure and volume. The primary difference between the EWCM and RWCM models lies in these additional terms. Equation 9 follows Kirchhoff's Current Law, which enforces mass conservation at junctions by ensuring that the algebraic sum of inflows and outflows equals the nodal demand. Presenting the system in this circuit form provides an intuitive physical understanding of how hydraulic energy is stored, dissipated, and transmitted through the network. By substituting Equations 7 and 8 into Equation 9, the following is obtained:

$$\left(\frac{1}{2} \sum_{j \in \Lambda_i} C_j\right) \frac{dh_i}{dt} = \sum_{j \in \Lambda_{i,d}} q_j - \sum_{j \in \Lambda_{i,u}} q_j + Q_i \quad (10)$$

2.1. Graph-Theoretical Formulation of WDSs

In a WDS, the state variables include the hydraulic head at each node (except reservoirs) and the flow rates in the pipes. By determining these unknown variables, the entire system's state can be identified.

This paper represents a WDS as a graph of pipes. In graph theory, a graph is represented by an incidence matrix, $\mathbf{M}$, where each row corresponds to a node, and each column represents an edge element. The matrix entries are either 1, -1 , or 0, depending on the relationships between nodes and edges. A value of 1 in $\mathbf{M}_{i,j}$ indicates that edge j exits node i , a value of -1 means edge j enters node i , and 0 signifies no connection between node i and edge j . Dividing nodes into internal and reservoir nodes (boundary nodes) allows us to partition the incidence matrix as:

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_I \\ \mathbf{M}_R \end{bmatrix} \quad (11)$$

where the subscript I refers to internal nodes and R to reservoir nodes. Similarly, the nodal head vector is partitioned as:

$$\mathbf{h} = \begin{bmatrix} \mathbf{h}_I \\ \mathbf{h}_R \end{bmatrix} \quad (12)$$

where $\mathbf{h}_I$ represents the heads at internal nodes, and $\mathbf{h}_R$ corresponds to the heads at reservoir nodes. Based on these partitions, the vector of head differences $\Delta \mathbf{h}$ across each element is expressed as:

$$\Delta \mathbf{h} = \mathbf{M} \mathbf{h} = \begin{bmatrix} \mathbf{M}_I \\ \mathbf{M}_R \end{bmatrix}^T \begin{bmatrix} \mathbf{h}_I \\ \mathbf{h}_R \end{bmatrix} \quad (13)$$

The head differences across pipes are derived from the incidence matrix and the nodal pressure head vector. As in Zeng et al. (2022), the demand vector $\mathbf{Q}$ and parameter matrices $\mathbf{L}$, $\mathbf{R}$, $\mathbf{C}$, and $\mathbf{T}$ (all diagonal) are employed in formulating the momentum and continuity equations for the pipes. The momentum equation is then written as:

$$\mathbf{L} \frac{d\mathbf{q}}{dt} = -\mathbf{R} \text{diag}\{|\mathbf{q}|\} \mathbf{q} + \mathbf{M}_I^T \mathbf{h}_I + \mathbf{M}_R^T \mathbf{h}_R \quad (14)$$

where $|\mathbf{x}|$ is the element-wise vector or matrix of $\mathbf{x}$. The continuity equation at the nodes is given by:

$$\frac{1}{2} \text{diag}\{|\mathbf{M}_I| \text{vec}\{\mathbf{C}\}\} \frac{d\mathbf{h}_I}{dt} = \mathbf{M}_I \mathbf{q} + \mathbf{Q} \quad (15)$$

where vec (Xing and Sela) is here defined as the vector of the diagonal elements of matrix $\mathbf{X}$. Equations 14 and 15 form a system of nonlinear ODEs for WDSs. This set of equations has been validated with MOC and the detail of the discretization of pipes can be found in the Zeng et al. (2022) and (Imani et al., 2025).

2.2. Nonlinear System Representation

The system of equations above can be compactly written in a state-space form. To do this, the state vector $\mathbf{x}$ is defined to include both the flow rates through the pipes and the hydraulic heads at the internal nodes:

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \mathbf{h}_I \end{bmatrix} \quad (16)$$

The inputs $\mathbf{u}$ to the system are the hydraulic heads at the reservoirs and the external demands:

$$\mathbf{u} = \begin{bmatrix} \mathbf{h}_R \\ \mathbf{Q} \end{bmatrix} \quad (17)$$

In this format, the system's nonlinear dynamics can be represented as:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \quad (18)$$

This equation describes how the state vector $\mathbf{x}$ evolves over time as a nonlinear function $\mathbf{f}(\mathbf{x}, \mathbf{u})$ of the state and the inputs. Solving this equation gives the complete transient response of the WDS to changes in demand or reservoir head, capturing both the flow rates and heads in the network.

3. Modal Analysis and Participation Factor

The following sections outline the key components of the proposed approach. First, the EWCM is linearized to obtain a state-space model suitable for modal analysis. Then, eigenvalue analysis and PF computations are performed to extract the system modes and assess their spatial characteristics. The AMM is also introduced in the frequency domain, and its methodology is compared with the mode extraction process based on the EWCM.

3.1. Linearization and State-Space Representation

To analyze the system using modal and frequency-domain methods, it's necessary to linearize the nonlinear system around an operating point. This is particularly useful for stability analysis, control design, and understanding the dominant modes of the system (Franklin et al., 2010). In practice, the operating point corresponds to the steady-state heads and flows obtained from a calibrated hydraulic model or from SCADA data under normal operating conditions. This state reflects the nominal behavior of the network. If alternative regimes (such as peak-demand or low-demand operation) are of interest, the same procedure can be repeated to obtain a family of linearized models that describe how modal properties vary across different operating conditions. Linearization involves approximating the nonlinear system with a linear one by expanding $\mathbf{f}(\mathbf{x}, \mathbf{u})$ in a Taylor series around the steady-state operating point $\mathbf{x}_0$ and $\mathbf{u}_0$. This assumes that flow and head perturbations are small compared to their nominal values. This yields the linearized state-space representation of the WDS:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (19)$$

where $\mathbf{A}$ and $\mathbf{B}$ are the Jacobian matrices of partial derivatives with respect to the state and input, respectively. Matrix $\mathbf{A}$ is the system matrix, which captures how the state variables (flow rates and hydraulic heads) change in response to perturbations in the state. Matrix $\mathbf{B}$ is the input matrix, which describes how external inputs (such as reservoir heads or demands) influence the state variables.

$$\mathbf{A} = \begin{pmatrix} \mathbf{L}^{-1}\mathbf{R}\text{diag}\{\mathbf{q}\} & \mathbf{L}^{-1}\mathbf{M}_I^T \\ 2\text{diag}\{\mathbf{M}_I|\text{vec}\{\mathbf{C}\}\}^{-1}\mathbf{M}_I & \mathbf{0}_{m \times m} \end{pmatrix} \quad (20)$$

where $\mathbf{0}_{m \times m}$ is an $m \times m$ matrix of zeros, and $\mathbf{L}^{-1}\mathbf{R}\text{diag}\{\mathbf{q}\}$ is clearly a diagonal matrix. Here, m denotes the number of internal nodes. The system matrix $\mathbf{A}$ is crucial for understanding the dynamic behavior of the WDS. Its eigenvalues determine the natural modes of the system, while the eigenvectors provide insight into how different components of the system interact with each mode.

3.2. Eigenvalue Analysis and Participation Factor

The eigenvalues λ_i of the system matrix A are found by solving the characteristic equation:

$$\det\{A - \lambda_i I\} = 0 \quad (21)$$

Each eigenvalue represents a mode of the system. If λ_i is real, the mode is non-oscillatory, and its sign determines whether the mode is stable ($\lambda_i < 0$) or unstable ($\lambda_i > 0$). Complex eigenvalues correspond to oscillatory modes, where the real part σ_i determines the stability and the imaginary part ω_i corresponds to the oscillation frequency:

$$\lambda_i = \sigma_i + i\omega_i \quad (22)$$

where i is the imaginary unit. For each eigenvalue, there are corresponding right and left eigenvectors, ν_i and w_i . The right eigenvector ν_i satisfies:

$$A\nu_i = \lambda_i\nu_i \quad (23)$$

and represents the mode shape, describing how the state variables (flow rates and heads) contribute to the dynamics of mode i . The left eigenvector w_i satisfies:

$$w_i^T A = \lambda_i w_i^T \quad (24)$$

and indicates how the mode i can be observed in the outputs. The PF p_{ki} for the k th state variable in the i th mode is defined as (Abdulrahman, 2020; Garofalo et al., 2002):

$$p_{ki} = \frac{\|v_{ki}\| \|w_{ki}\|}{\sum_{k=1}^n \|v_{ki}\| \|w_{ki}\|} \quad (25)$$

$$\bar{p}_{ki} = \frac{p_{ki}}{\max_{i=1,\dots,n} |p_{ki}|} \quad (26)$$

where $\|\cdot\|$ denotes the L_1 -norm operator, n is the total number of state variables, and $\bar{p}_{ki}$ is the normalized PF. These factors quantify how much the k th state variable contributes to the i th mode, combining information from both the left and right eigenvectors. PFs have key properties that make them valuable for system analysis. First, the sum of PFs for each mode across all state variables equals 1, meaning all variables together represent the full dynamics of that mode. Similarly, the sum for each state variable across all modes also equals 1, indicating the state variable contributes fully to the system's overall behavior (Abed et al., 2000). Additionally, PFs reveal how sensitive eigenvalues are to changes in system parameters, helping to identify critical components for control and design decisions.

3.3. Admittance Matrix Method for Mode Extraction

In the AMM, the dynamic behavior of a WDS is analyzed in the Laplace domain. The detailed methodology can be found in (Zecchin, 2010). This method applies the Laplace transform to the continuity and momentum equations to convert the equations from the time domain to the frequency domain with the Laplace operator s . The Laplace operator $s = \sigma + i\omega$, where σ represents the decay rate, and ω represents the frequency. The WDS is then modeled using an admittance matrix $Y(s)$, which relates the nodal heads $h(s)$ (the Laplace transform of h) and flows $Q(s)$ (the Laplace transform of the demands and reservoir flows) at each node. The admittance matrix is constructed as follows (Zecchin et al., 2018):

$$Y(s)h(s) = Q(s) \quad (27)$$

$$Y(s) = \begin{pmatrix} N_u & N_d \end{pmatrix} \begin{pmatrix} Z_c^{-1}(s) \coth \Gamma(s) & -Z_c^{-1}(s) \operatorname{csch} \Gamma(s) \\ -Z_c^{-1}(s) \operatorname{csch} \Gamma(s) & Z_c^{-1}(s) \coth \Gamma(s) \end{pmatrix} \begin{pmatrix} N_u & N_d \end{pmatrix}^T \quad (28)$$

Where N_u and N_d are the upstream and downstream incidence matrices, respectively, and contain the information regarding the topology of the system. These matrices have the following relation with the incidence matrix from the previous sections:

$$N_u = \frac{\mathbf{M} + |\mathbf{M}|}{2} \quad (29)$$

$$N_d = \frac{|\mathbf{M}| - \mathbf{M}}{2} \quad (30)$$

In Equation 27, $\mathbf{\Gamma}(s)$ is the propagation operator, and $\mathbf{Z}_c(s)$ is the characteristic impedance (see (Zecchin et al., 2009) for details).

By partitioning the system into controlled nodes (inputs) and free nodes (outputs), the network matrix Equation 27 can be rewritten in the following partitioned form (Zecchin et al., 2009):

$$\begin{pmatrix} \mathbf{Y}_d(s) & \mathbf{Y}_{d-r}(s) \\ \mathbf{Y}_{r-d}(s) & \mathbf{Y}_r(s) \end{pmatrix} \begin{pmatrix} \mathbf{h}_I(s) \\ \mathbf{h}_R(s) \end{pmatrix} = \begin{pmatrix} \mathbf{Q}(s) \\ \mathbf{Q}_R(s) \end{pmatrix} \quad (31)$$

where $\mathbf{Q}_R$ are the Laplace transform of reservoir flows (positive flow is into the network), $\mathbf{Y}_d(s)$ is the $n_d \times n_d$ system matrix representing the demand nodes, and $\mathbf{Y}_r(s)$ is the $n_d \times n_d$ system matrix representing the reservoir nodes. $\mathbf{Y}_{d-r}(s)$ and $\mathbf{Y}_{r-d}(s)$ are the respective partitions of the network matrix that describe the contributions of flow at the demand nodes and reservoir nodes, resulting from the nodal heads at the opposite set of nodes. It is important to note that $\mathbf{Y}_d(s)$ and $\mathbf{Y}_r(s)$ are symmetric, and $\mathbf{Y}_{d-r}(s) = \mathbf{Y}_{r-d}^T(s)$.

The transfer function is then derived based on this partitioning. Zecchin et al. (2009) formulated expressions for the input-output (I/O) transfer function as follows.

$$\mathbf{H}(s) = \begin{pmatrix} \mathbf{H}_{11}(s) & \mathbf{H}_{12}(s) \\ \mathbf{H}_{21}(s) & \mathbf{H}_{22}(s) \end{pmatrix} = \begin{pmatrix} -\mathbf{Y}_d^{-1}(s)\mathbf{Y}_{d-r}(s) & \mathbf{Y}_d^{-1}(s) \\ \mathbf{Y}_r(s) - \mathbf{Y}_{r-d}(s)\mathbf{Y}_d^{-1}(s)\mathbf{Y}_{d-r}(s) & (\mathbf{Y}_d^{-1}(s)\mathbf{Y}_{d-r}(s))^T \end{pmatrix} \quad (32)$$

For modal analysis, the system's transfer function is expressed in a mode-based form, where the system's dynamics are described by a sum of modal terms (Zecchin et al., 2018):

$$G(s) = \sum_{n=1}^N \left(\frac{G_n}{s - z_n} + \frac{G_n^*}{s - z_n^*} \right) \quad (33)$$

where N represents the number of modes, G_n is the complex-valued modal coefficient matrix for the n th mode, and $z_n = \sigma_n + i\omega_n$ is the complex frequency for the n th mode. The operator $*$ denotes the complex conjugate.

To compute the modes, the poles of the transfer function $G(s)$ must be located in the complex plane. These complex modal frequencies correspond to the singular points of G , where $\{z \in \mathbb{C} : |G_{(z)}| = 1\}$. To determine the complex frequencies z_n , it is necessary to identify the locations of the poles in the analytic I/O transfer function $H(s)$.

The general process of finding the complex frequencies z_n involves determining bounds for their real and imaginary components (i.e., $\sigma_n \in [\sigma_{\min}, \sigma_{\max}]$ and $\omega_n \in [\omega_{\min}, \omega_{\max}]$), followed by locating z_n by identifying the maxima of the elements of $|H|$ within these bounds, which correspond to the pole locations. The process for finding the pole locations can be found in (Zecchin et al., 2018).

3.4. Comparison of AMM and EWCM in Modal Analysis

Modal information can, in principle, be obtained using several classical approaches. In the time domain, the MOC can be used by exciting the system, recording the pressure and flow responses at discrete nodes, and performing Fourier or Laplace transforms to construct frequency-response functions. The locations of the poles in the Laplace

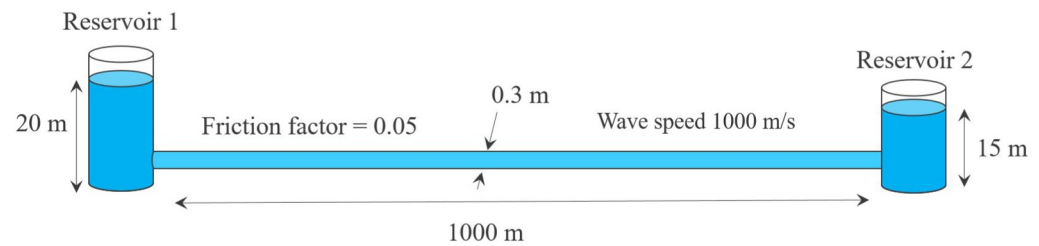


Figure 2. Structure of the pipeline for test case 1.

plane can then be inferred from the resonance peaks, poles located closer to the imaginary axis correspond to larger amplitude responses. While this approach is conceptually straightforward, it becomes computationally demanding for large-scale networks, as multiple transient simulations are required to capture the full spatial response.

Similarly, the Transfer Matrix Method (TMM), which is formulated in the frequency domain, allows the transfer function of a single pipe or a simple series system to be constructed and its FRD obtained, from which the poles can also be inferred. However, the traditional TMM formulation is primarily developed for single-pipe systems, and extending it to general network configurations requires additional derivations. For this reason, the AMM is used in this study as the benchmark for comparison. The AMM is formulated in the Laplace domain, and an established framework exists for extracting poles for arbitrary network topologies.

The AMM and the EWCM both offer distinct advantages in the modal analysis of water distribution systems, depending on the context and objectives of the analysis. The AMM is efficient for analyzing the system's input–output relationships and allows quick identification of resonance frequencies by constructing the frequency–response function. Extracting modes from the AMM, however, requires specific steps such as sweeping through a frequency range and locating poles, which can involve additional computational effort. Despite this, it remains a reliable approach for understanding a system's dynamic characteristics in the frequency domain.

The EWCM, in contrast, operates in the time domain and provides a model based on ODEs that approximate the underlying PDEs of unsteady flow. When linearized about a steady operating point, the EWCM yields a state-space representation in which modal information follows directly from the eigenvalues and eigenvectors of the system matrix. This provides direct access to both right and left eigenvectors, enabling the computation of PFs and detailed spatial interpretation of each mode, capabilities that are not directly available from frequency-domain approaches such as the AMM. These distinctions reflect the classical difference between transfer-function models and state-space models described in control theory texts (e.g. (Franklin et al., 2010)): transfer-function formulations capture input–output behavior, while state-space models also expose the internal dynamics of the system.

4. Numerical Simulation

In this section, three test cases are presented to demonstrate the application of the modal analysis and PF analysis framework developed for WDS using the EWCM. The first test case focuses on a single pipeline to explore the fundamental behaviors and validate the extracted modes using both the EWCM and the AMM. The second test case extends the analysis to a pipe network, introducing complexities such as loops and multiple internal nodes. For these two test case, eigenvalue analysis and PF analysis are employed to extract the system modes and assess the contributions of different system states to each mode. Time-domain simulations are used to further verify the results. In Test Case 3, a real world test case, New York tunnel has been chosen to show the applicability of PF in revealing critical locations due to hydraulic transients.

4.1. Test Case 1: Single Pipeline

In this test case, the modal analysis for a single pipeline, as shown in Figure 2, is presented.

To begin with, the EWCM is used to determine the modes, where the spatial discretization is governed by the critical frequency ($f_c = a/10l$). The segment length l is chosen such that the highest frequency of interest satisfies $f_{\max} \leq f_c$. In this test case, a maximum frequency of 10 rad/s was considered; using this value, the corresponding

Table 1
Eigenvalue of System Matrix for Test Case 1

Number	Eigenvalue	Number	Eigenvalue
1	-0.1278	16,17	$-0.0639 \pm 22.627i$
2,3	$-0.0639 \pm 3.135i$	18,19	$-0.0639 \pm 24.736i$
4,5	$-0.0639 \pm 6.242i$	20,21	$-0.0639 \pm 26.606i$
6,7	$-0.0639 \pm 9.288i$	22,23	$-0.0639 \pm 28.221i$
8,9	$-0.0639 \pm 12.245i$	24,25	$-0.0639 \pm 29.564i$
10,11	$-0.0639 \pm 15.084i$	26,27	$-0.0639 \pm 30.622i$
12,13	$-0.0639 \pm 17.778i$	28,29	$-0.0639 \pm 31.385i$
14,15	$-0.0639 \pm 20.300i$	30,31	$-0.0639 \pm 31.845i$

critical frequency was applied to determine the appropriate pipe-segment length and, consequently, the number of reaches for each pipe. As a result, the pipeline was discretized into 16 reaches. Based on this discretization, the system matrix has a dimension of 31×31 , where 16 states correspond to flow rates and 15 states correspond to the pressure heads at internal nodes. Performing eigenvalue analysis on this matrix yields 31 eigenvalues, which are listed in Table 1.

As observed in Table 1, there is one pole with no imaginary part, representing a damped mode without an oscillatory component. This damped pole can also be found using the RWCM. For a single pipeline with a fixed head, the equation for this pole is given by $\frac{fa|q_0|}{IDS}$, and the detailed derivation for obtaining this damped mode through the RWCM is provided in Appendix A.

The remaining poles have both real and imaginary parts. For this single pipeline, all oscillatory modes have equal real parts and are complex conjugates, meaning that their decay rates are identical, with differences only in their imaginary components. The real part of the poles is influenced by the system's resistance, while the fluid compressibility modeled as capacitance in EWCM impacts the imaginary part.

To verify the pole locations, the mode extraction procedure using the AMM was also applied to this pipeline. The transfer function surface is shown in Figure 3, where the local maxima indicate the locations of the poles. By performing a 2D optimization, the exact pole locations were extracted, as shown in Figure 4. It is important to note that only the positive frequencies are displayed in Figures 3 and 4 for clarity. A comparison of the pole locations obtained by both methods reveals that the poles from the EWCM match those from the AMM up to the critical frequency. Beyond this frequency, the EWCM's poles diverge from the actual poles. However, by increasing the discretization, the bandwidth can be expanded, allowing the EWCM to capture higher-frequency modes. Additionally, the location of the damped mode is in agreement across all three models EWCM, RWCM, and the AMM.

By obtaining the eigenvalues along with the left and right eigenvectors, the PF matrix can be constructed for this test case. In this matrix, the rows correspond to the system states, and the columns represent the modes. Since only modes 1 to 4 are valid (represented in Figure 4), the PFs for these modes are extracted and shown in Figure 5. A PF of 1 indicates the highest participation of a state in a given mode, while a value of 0 signifies no participation. Mode 1 represents the damped mode, and only the flow state variables contribute to this mode, with all variables having equal importance. This result can be verified using the RWCM, as in this model, the left-hand side of Equation 10 is zero, meaning there is no partial derivative of the internal head over time. For the other oscillatory modes, the PFs are illustrated in Figure 6 (only for modes with positive imaginary parts).

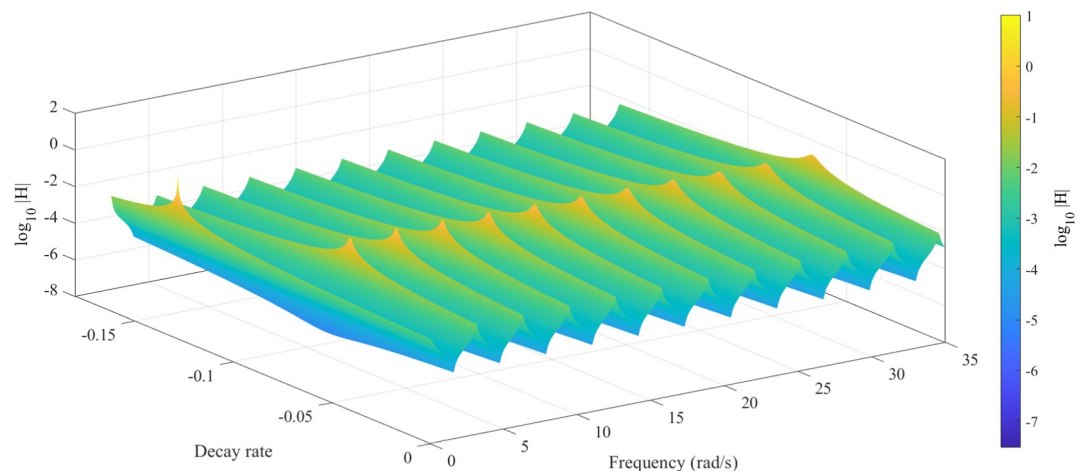


Figure 3. The surface of the absolute value of the Laplace-domain response surface for test case 1.

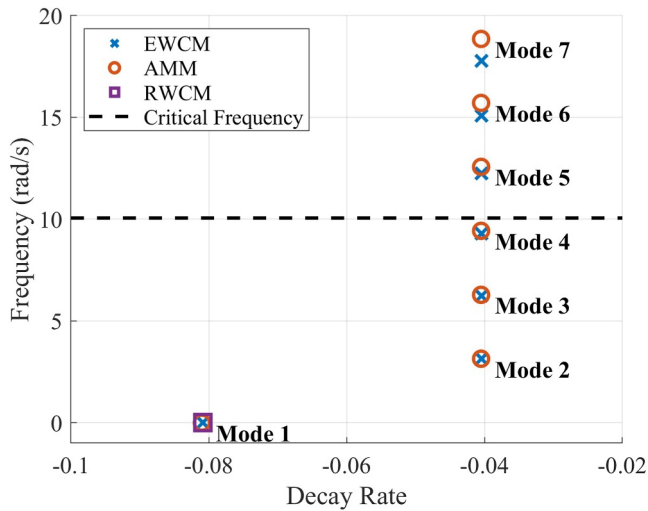


Figure 4. Locations in the Laplace domain for Test Case 1.

In Figure 6, the modes shown correspond to the first three harmonics of the system. For the mode 2, the PF for the internal head variable is highest at the center of the pipe. This behavior can be understood by drawing an analogy to systems such as a mass-spring-damper or a vibrating string fixed at both ends. In the first harmonic of the system, there is an antinode at the center of the pipe, where the potential energy is maximized, similar to the maximum displacement in a vibrating string. The two fixed heads at the pipe's input and output act as nodes, where the displacement is zero. This pattern can be extended to the higher harmonics, represented by modes 2 and 3 (the second and third harmonics of the pipeline). In these modes, additional nodes and antinodes appear along the length of the pipe, with the internal head variable participating more significantly at certain points, depending on the mode shape. These harmonics have been explained in detail by (Wylie & Streeter, 1993).

To verify the PF matrix and identify the most sensitive locations for each mode, a time-domain simulation was performed. The system was excited at the location with the highest PF, with the excitation frequency matching the mode frequency. To excite the system at point 6 (Node numbering in

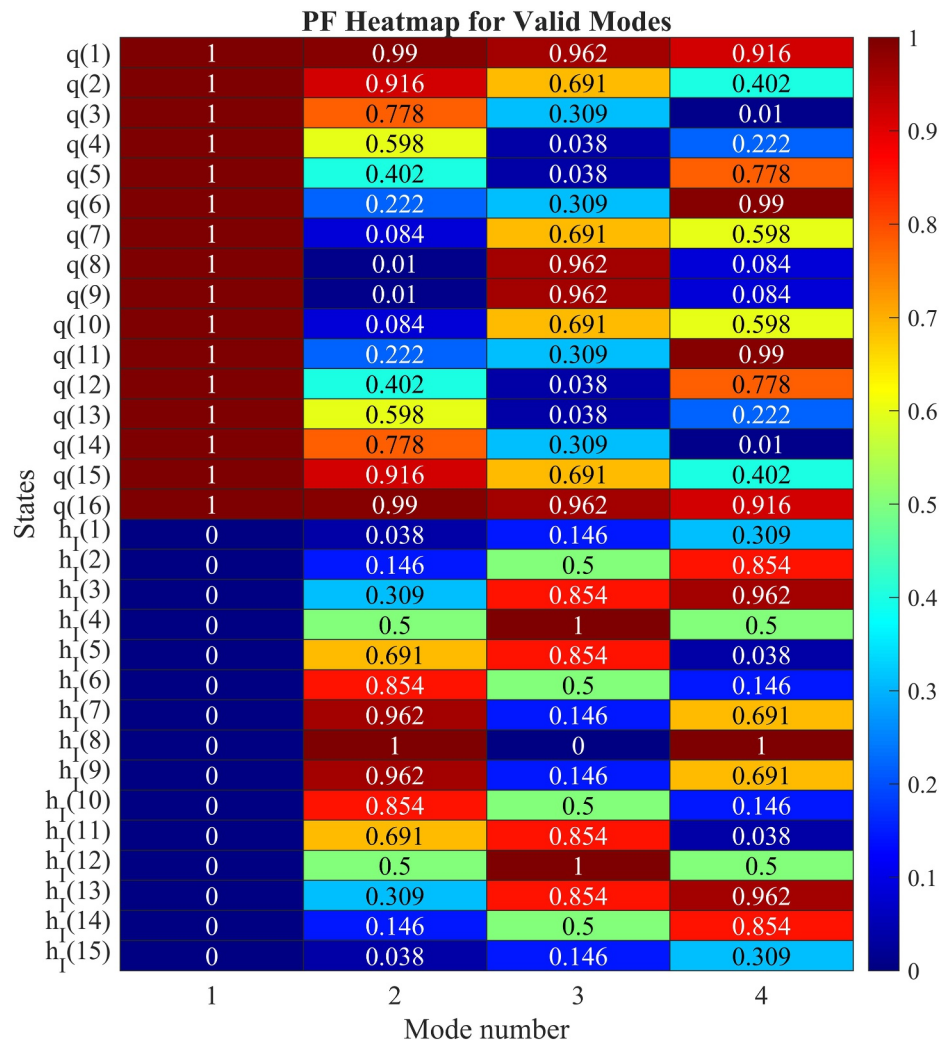


Figure 5. Participation matrix for valid modes.

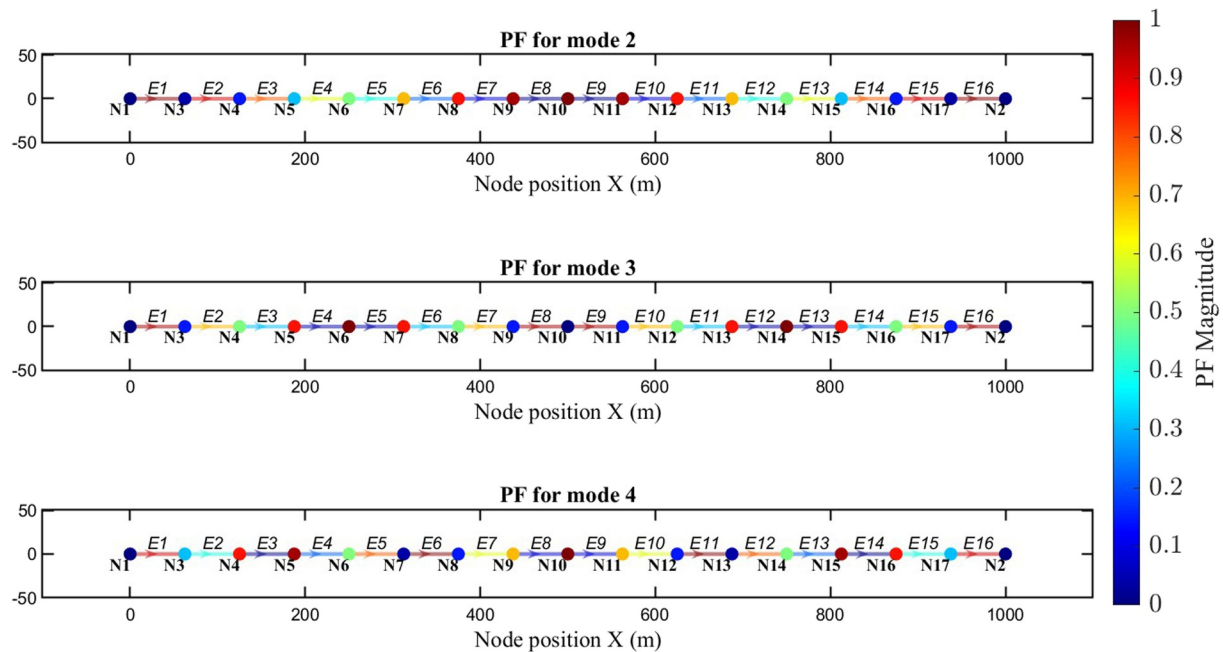


Figure 6. Representation of PFs on the pipeline: nodes are associated with participation of nodal heads, and edges are associated with flow rates through pipes.

Figure 6), which exhibits the highest PF for the second oscillatory mode ($-0.0639 \pm 6.242i$), an oscillatory demand, given by the equation below, was used.

$$Q(t) = 0.01 + 0.0005 \times \sin(6.242t) \quad (34)$$

In Equation 34, the constant term ($0.01 \text{ m}^3/\text{s}$) represents the steady-state demand, while the oscillatory term ($0.0005 \text{ m}^3/\text{s} = 0.5 \text{ L/s}$) was chosen as 5% of the nominal demand. This amplitude ensures that the excitation remains within the linear range of the model while later allowing the analysis to demonstrate the system's sensitivity, where even such a small perturbation can lead to noticeable pressure variations depending on the excitation location. The simulation results, after reaching a steady oscillatory state, are shown in Figure 7, which

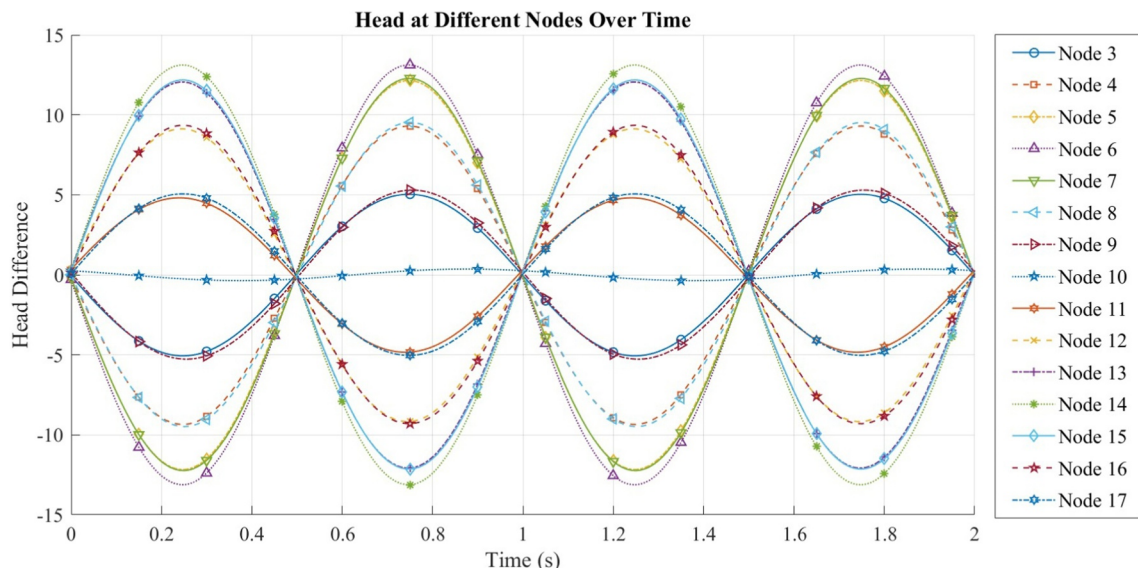


Figure 7. Head changes at all internal nodes under steady oscillatory conditions, with the excitation point at node 6.

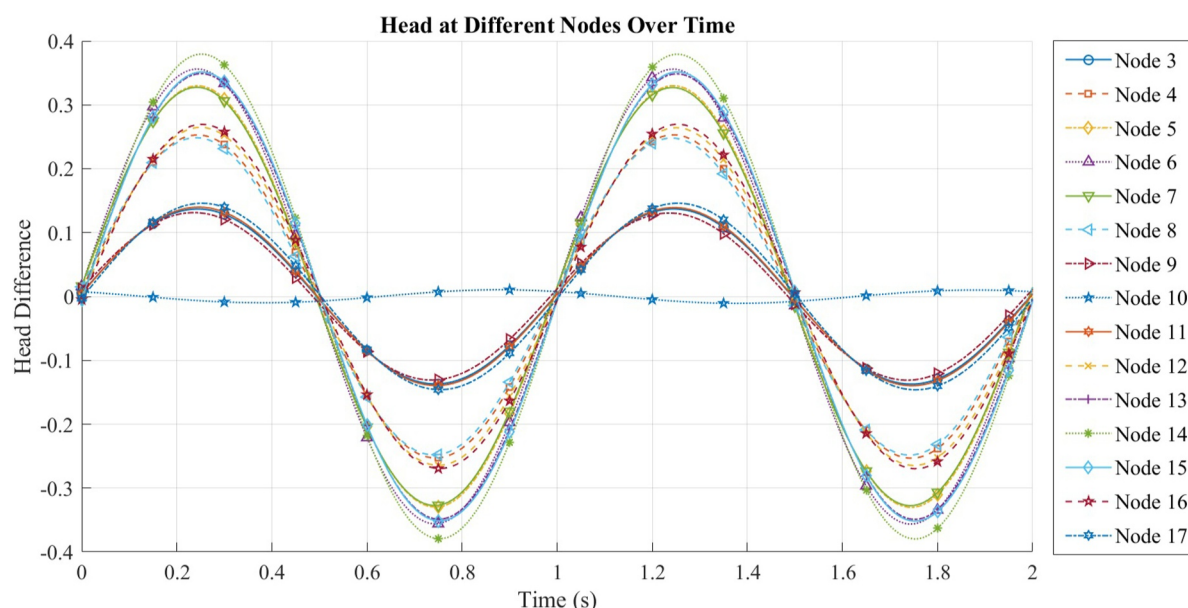


Figure 8. Head changes at all internal nodes under steady oscillatory conditions, with the excitation point at node 10.

illustrates the variation of pressure at internal nodes ($h(t) - h_0$). It can be seen that a small variation in demand (0.5 L/s) causes a head change of approximately 10 m at nodes 6 and 14, which correspond to the locations with the highest PF values. From Figure 6, node 10 has the lowest PF for the second oscillatory mode, and the time-domain simulation in Figure 7 confirms that this node exhibits the smallest oscillation amplitude under steady oscillatory conditions.

To further demonstrate the influence of excitation location, the demand oscillation was next applied at the lowest-PF node (node 10) for the same mode, using the same excitation defined in Equation 34. The resulting head variations, shown in Figure 8, indicate that the maximum head change was less than 0.4 m, significantly smaller than the response observed when exciting at the high-PF location. This comparison shows that identical excitations can lead to markedly different dynamic responses depending on the spatial location of the applied disturbance. When the excitation is applied at a high-PF node, the pressure response is amplified due to resonance

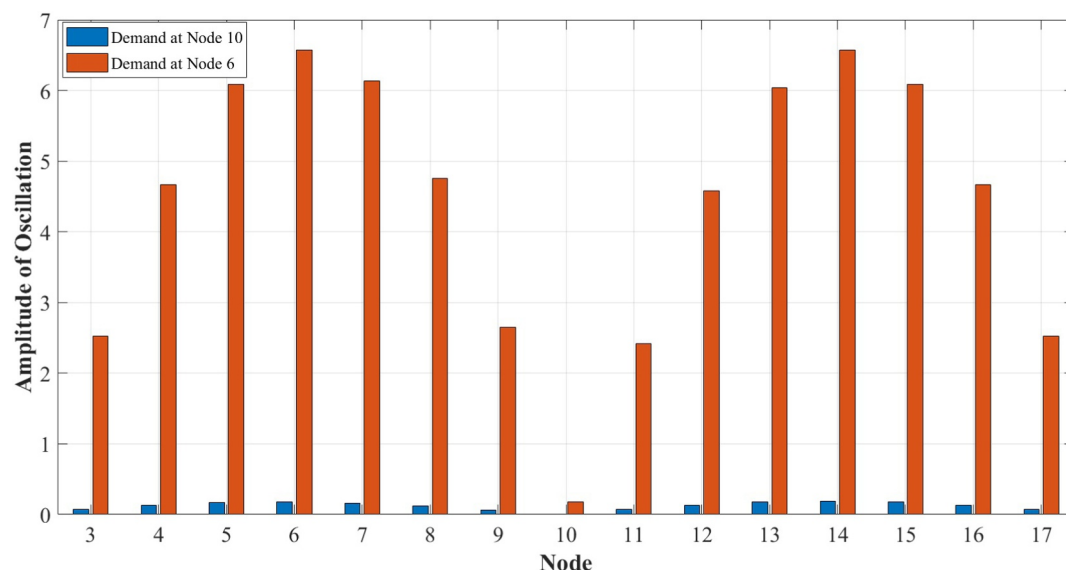


Figure 9. Fast Fourier Transform of internal node heads, with the excitation point at node 6 and 10.

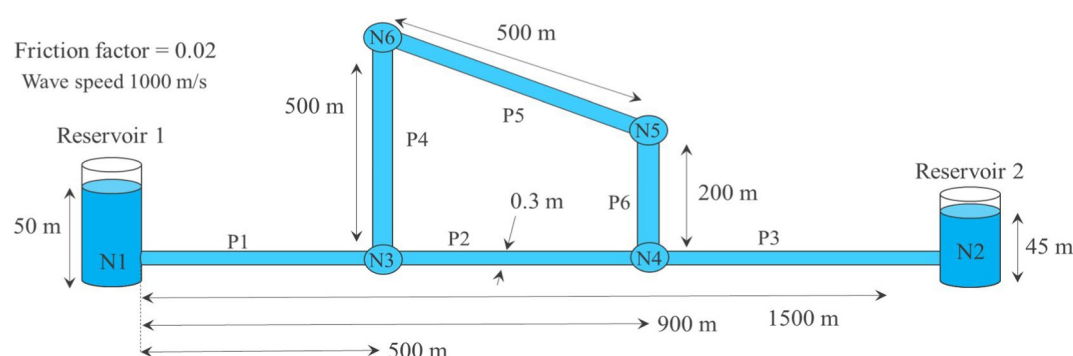


Figure 10. Network structure for Test Case 2.

with the corresponding mode, whereas excitation at a low-PF node produces only a weak response that is rapidly attenuated. The spatial pattern of the simulated head amplitudes in Figure 9 closely matches the PF distribution in Figure 6, illustrating that the regions with higher PF values experience greater oscillation amplitudes under modal excitation. This agreement verifies that the PFs capture the underlying spatial characteristics of the oscillatory modes observed in the time-domain simulation.

4.2. Test Case 2: Pipe Network

In this test case, the modal analysis for the pipe network shown in Figure 10 is implemented. The system contains one loop, where all the details are provided in Figure 10. In the first step, the pole locations were identified using both the EWCM and the AMM, as shown in Figure 11. The number of reaches in the system was chosen so that there is one reach every 100 m in the system. Therefore, the number of reaches for the pipes is 5, 4, 6, 5, 5, and 2 respectively.

In Figure 11, by neglecting the poles that lie on the critical frequency line, there are only four oscillation modes below the critical frequency. In this system, there are two damped modes, were also identified by the RWCM. According to (Shimada, 1992), the number of independent states is equal to the difference between the number of pipes and the number of internal nodes. In this case, the system has 6 pipes and 4 internal nodes, resulting in two negative real poles. These damped modes are associated with the loop and the spanning tree between the two reservoirs. As can be seen, all poles below the critical frequency align closely for both the EWCM and AMM. A slight deviation is observed for Mode 4, for which the relative error is below 1%. This small difference arises from the grid resolution and the optimization procedure used in the AMM to identify modal poles.

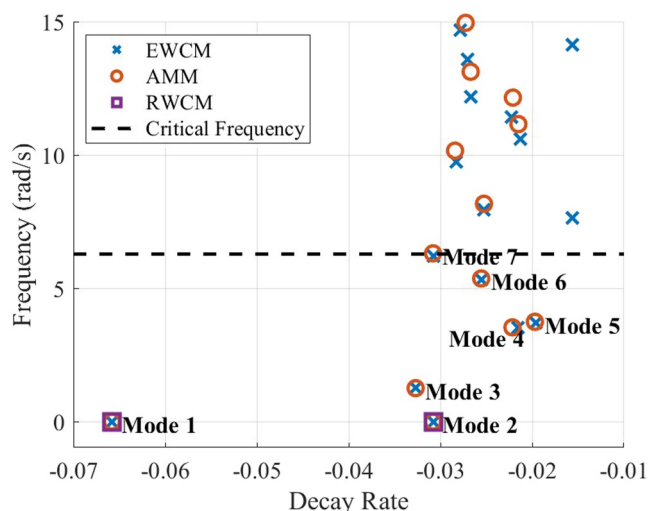


Figure 11. Pole locations for Test Case 2.

The 3D surface of the transfer function in the Laplace domain, obtained using the AMM, is shown in Figure 12. The line with a decay rate equal to zero represents the FRD of the system between the input and output. The location of the poles affects the FRD of the system.

Figure 13 shows the PFs for all valid modes. As seen in Figures 13a and 13b, for the two damped modes, only the flow rate states participate in the modes, as indicated by the dark blue color in the links representing flow variables. The flow rate states of Pipe 2 participate in both damped modes.

For the oscillatory modes, Mode 3 ($-0.032 + 1.263i$), shown in Figure 13c, behaves like the first harmonic of the network. This means there is one anti-node (Node 6) with maximum potential energy (similar to a mass-spring-damper system) and two nodes with maximum kinetic energy (Node 1 and 2). Modes 4 ($-0.021 + 3.52i$) and Mode 5 ($-0.019 + 3.71i$) should be considered together since they form the second harmonic of the system and are closely related. When both modes are examined simultaneously, it is evident that nodes N3 and N4 serve as anti-nodes for the second harmonic and are the most sensitive locations, experiencing the most oscillation during the excitation of these modes. For Mode 4, Pipe 1 shows no participation, while

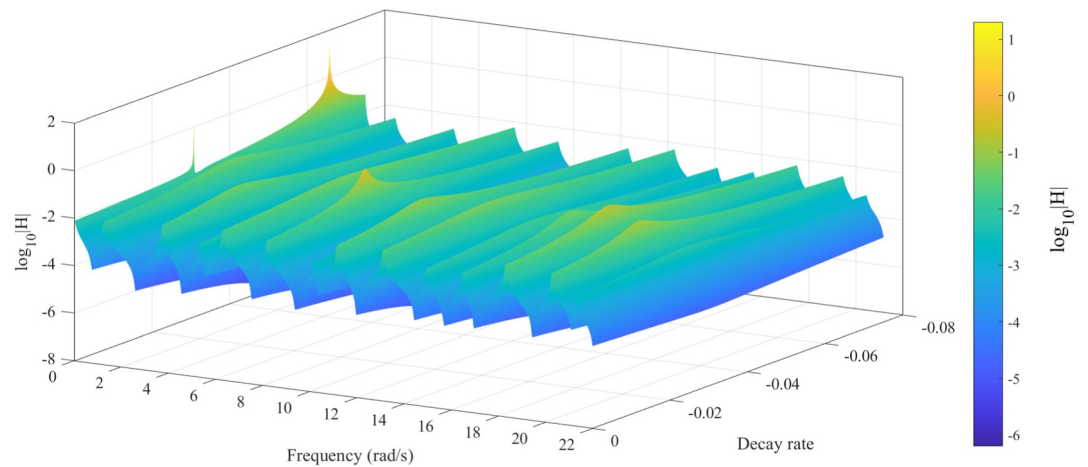


Figure 12. Surface plot of the Laplace-domain response for Test Case 2.

for Mode 5, Pipe 6 shows no participation. For the nodes on the pipe loop, the participation alternates: if a node has participation in either Mode 4 or 5, it shows little or no participation in the other. Among them, only Node 23 participates in both modes, with PFs of 0.38 and 0.47 for modes 4 and 5, respectively.

By conducting a time-domain simulation and placing an oscillating demand at Node 23, as in Test Case 1, the system can be excited. The results of this simulation are shown in Figure 14. For both frequencies, the system exhibits significant head variation. For excitation with the frequency of Mode 4 (3.5 rad/s), the maximum head change is 10 m, while for excitation with frequency of Mode 5 (3.71 rad/s), the maximum head change is 13.5 m. These are considerable head changes for a demand variation of only 0.5 L/s. Although the oscillation frequencies for these two modes are close, they demonstrate distinct behaviors. From Figure 14, it is evident that node 16 shows no participation in Mode 5 but experiences a 10-m head change during oscillation in mode 4.

4.3. Test Case 3: The New York Tunnel System

In this test case, the practicality of PF Analysis for WDSs has been investigated. The New York tunnel system has been extensively studied by numerous researchers for both steady-state and transient analysis (Dandy

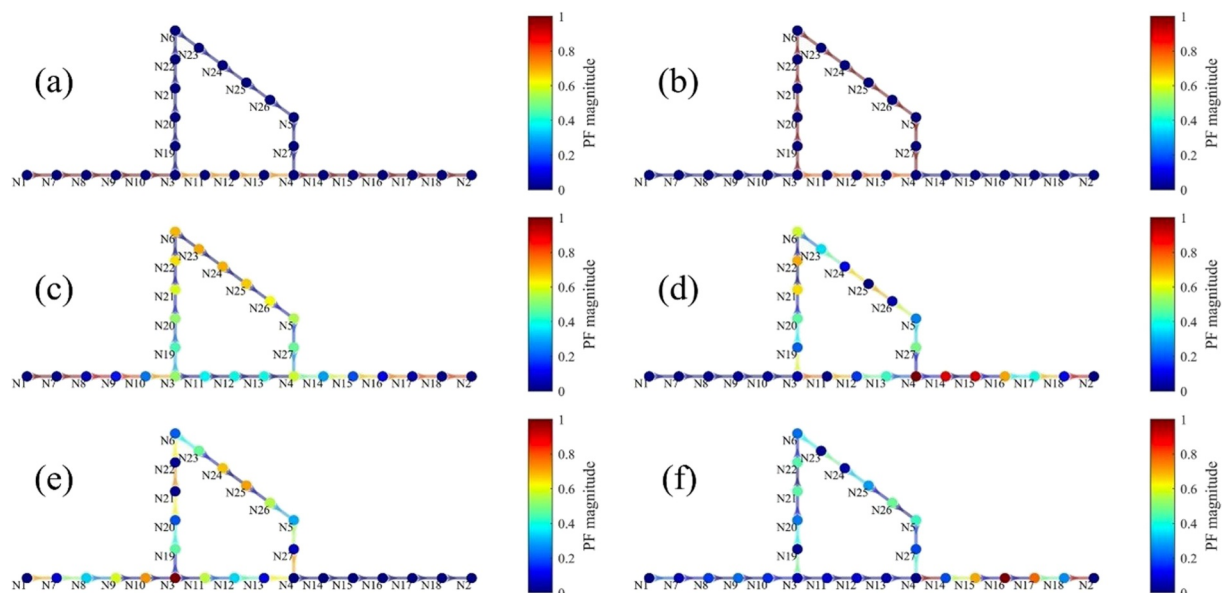


Figure 13. Representation of participation factor on the network for all valid modes: (a) Mode 1, (b) Mode 2, (c) Mode 3, (d) Mode 4, (e) Mode 5, and (f) Mode 6.

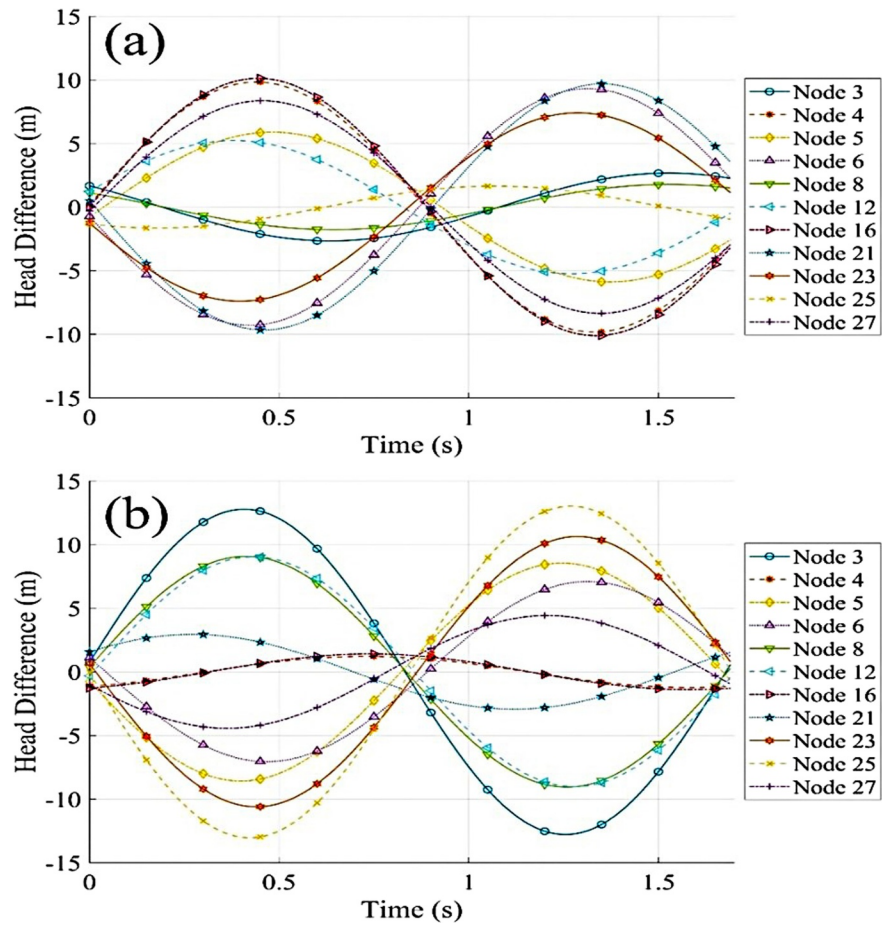


Figure 14. Head change at all internal nodes under steady oscillatory conditions with excitation applied at node 10: (a) oscillation frequency of 3.5 rad/s; (b) oscillation frequency of 3.7 rad/s.

et al., 1996). One approach to incorporating transient analysis into the design was to identify the worst-case scenario and apply optimization techniques to reduce transient effects (Jung Bong & Karney Bryan, 2009; Jung et al., 2011).

Details of the system can be found in (Dandy et al., 1996). It was determined that nodes 17 and 19, which are dead ends, are more critical compared to other nodes and can generate high transients (Jung, 2022).

In the previous test cases, the system was excited using only a single frequency. However, in real-world scenarios, system excitation consists of a combination of frequencies. Figure 15 presents a demand increase over 2 s, along

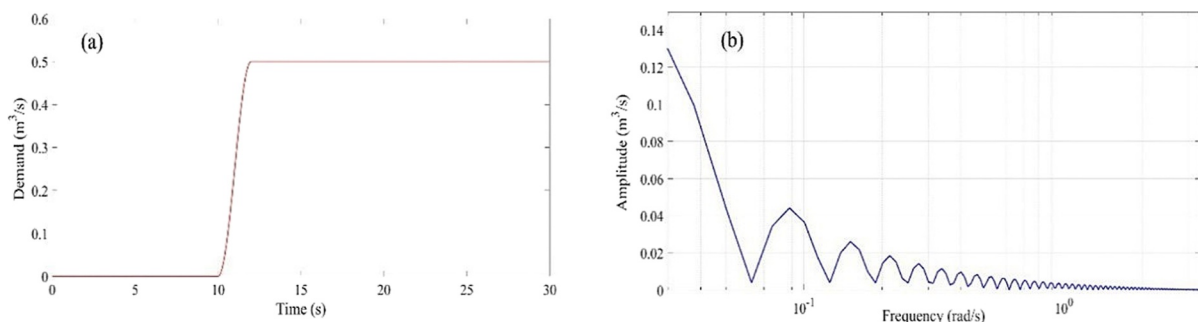


Figure 15. Generated hydraulic transient by increasing demand. The rise follows a sine pattern to smooth the transition: (a) signal in time domain; (b) frequency content of the signal.

Weighted Average PF for Modes Below f_c

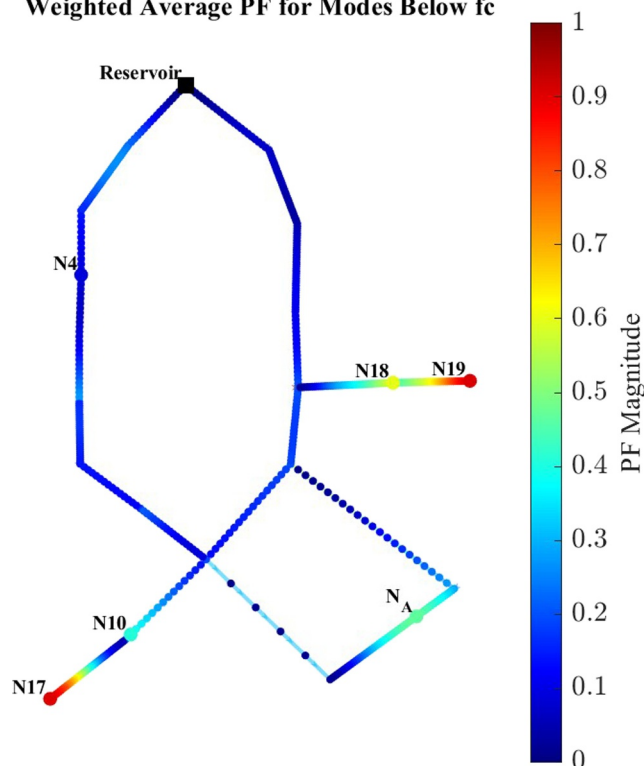


Figure 16. Weighted average participation factor up to the critical frequency.

with its frequency spectrum. The demand rises from 0 to 0.5 m³/s following a sine-shaped pattern to ensure a gradual transition and avoid introducing artificial high-frequency components. As shown, low frequencies exhibit higher amplitudes compared to high frequencies. This frequency amplitude can be utilized in PF analysis. After identifying the modes, a weighted average of PFs up to the critical frequency can be computed using the frequency amplitude as the weighting factor. Specifically, the amplitudes of each frequency component obtained from the Fourier spectrum (Figure 15b) were used to weight the normalized participation factors defined in Equation 26. The weighted average was then calculated as the sum of each mode's PF multiplied by its normalized frequency amplitude, divided by the sum of all amplitudes, ensuring that more strongly excited frequencies contribute proportionally to the overall PF distribution. The resulting averaged PF provides an overall representation of the state contributions to the system's transient response and helps identify critical locations.

Figure 16 shows the weighted average PF up to the critical frequency. The system has been discretized such that a reach is defined every 200 m. As seen in Figure 16, the dead ends (nodes 17 and 19) are the most critical nodes. Node N_A , which is also shown in the graph, exhibits a high PF. This node has not been identified in previous studies. Node 10 also exhibits a high PF, which can be attributed to a significant impedance mismatch at this location. The impedance of a pipe is defined as $B = \sqrt{L/C} = a/gS$. An impedance mismatch occurs when two pipes with different diameters and/or materials (which influence wave speed) are joined together. Such mismatches are well-known in power systems and fluid dynamics for causing partial reflection and transmission of transient waves (Bohorquez et al., 2020; Gong et al., 2013, 2018; Zeng et al., 2018). In this case, the sudden change in pipe diameter near

Node 10 creates a discontinuity in impedance. When a transient wave generated in a larger-diameter pipe reaches this node, part of the energy is reflected while the rest is transmitted with distortion. This interaction excites multiple modes in the system and increases the dynamic activity at this location, which is why Node 10 appears as a region with high PF in the analysis. To verify the results, a transient, as shown in Figure 15 has been applied to the specified nodes, and the system response is presented in Figure 17. The transient behavior of systems with impedance mismatches has been investigated in previous studies, and interested readers are referred to those works for further details (Bohorquez et al., 2020; Gong et al., 2018).

Figure 17 shows the results of a demand increase at nodes 4, 10, N_A , 17, 18, and 19. Node 4 was chosen to demonstrate the system response when the excitation point is a low-PF region. The other nodes, as can be seen, generate high transients in the system according to their PF analysis. By using this approach, intermediate nodes along the pipes can also be investigated, and the accuracy can be increased by increasing the number of discretizations. This approach can provide a clear picture of the vulnerable areas of a WDS to pressure surges and hydraulic transients. The results confirm that exciting the system at nodes 17 and 19 generates high transients.

5. Practical Considerations, Uncertainties, and Future Directions

The numerical examples illustrate that modal analysis can reveal the underlying dynamic structure of a WDS in a way analogous to its use in other engineering disciplines such as structural or power-system dynamics. As in those fields, the method provides a means of decomposing complex behavior into a small number of characteristic modes, which can assist in understanding and managing system response. However, several practical considerations and limitations should be acknowledged, as outlined below.

First, the framework relies on linearization around a steady operating point. In practice, a network may operate under multiple demand conditions and flow regimes, each with slightly different hydraulic behavior. The analysis can therefore be repeated for several representative operating points, such as low-, average-, and high-demand scenarios to obtain a family of modal descriptions. These can be interpreted together to understand how the dominant modes evolve across normal operating conditions. Similarly, the level of spatial discretization

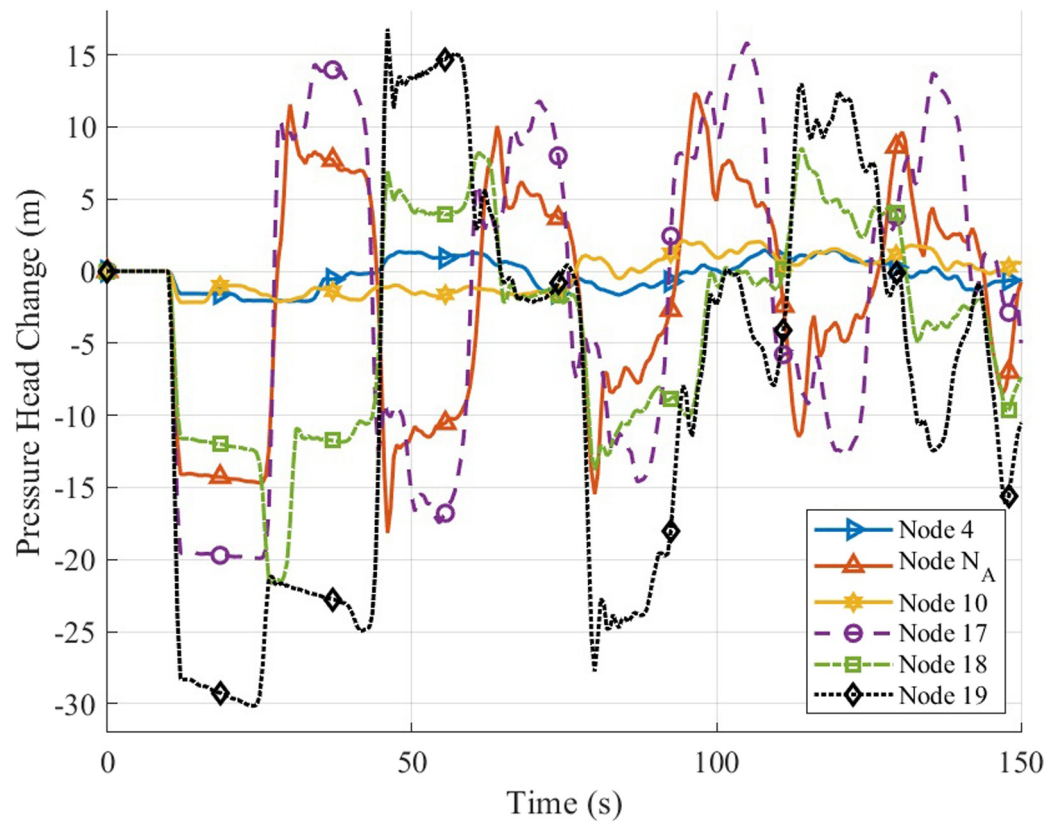


Figure 17. Transient response of the system at excitation points.

determines the highest frequency that can be resolved: finer discretization increases accuracy but also computational cost. The appropriate level depends on the purpose of the study, whether the aim is general behavioral insight, surge assessment, or preparation of reduced-order models for control applications.

Second, uncertainties in model parameters such as friction factors, wave speeds, and demand estimates are unavoidable in practice. Experience from related disciplines suggests that such variations shift modal frequencies and damping ratios slightly but rarely change the qualitative pattern of the modes. These uncertainties can be examined through sensitivity analysis or Monte-Carlo sampling to establish confidence intervals for the dominant eigenvalues and to ensure that the identified modes remain physically meaningful. In most cases, the participation-factor distributions are especially robust, because they represent relative spatial contributions rather than absolute magnitudes.

Third, applying the method to large networks introduces computational challenges, as the system matrices grow rapidly with the number of pipes and nodes. Nonetheless, the EWCM formulation leads to sparse matrices, where only adjacent elements interact (Imani et al., 2025). Modern sparse eigensolvers can efficiently extract the few lightly damped modes of engineering interest, making analysis of networks with hundreds of pipes tractable. For very large systems, one practical approach is to focus on critical sub-networks or to use the identified dominant modes to construct reduced-order models that preserve essential dynamics while lowering computational cost.

Future work may extend the present framework in several directions. Reduced-order models derived from dominant modes could be coupled with real-time monitoring data to enable faster transient prediction or adaptive control. Incorporating parameter uncertainty directly into the eigen-analysis, for example, through stochastic or interval methods, could further clarify the robustness of modal indicators. Integration with control design and data assimilation frameworks may ultimately allow the method to support operational decision-making, such as identifying excitation sources, refining surge protection strategies, or evaluating the impact of system modifications.

6. Conclusion

This paper demonstrates the use of the EWCM for modal analysis of water distribution systems WDSs. By linearizing the EWCM around an operating point, a state-space representation is established, allowing the extraction of natural modes through eigenvalue analysis. The results are verified against the AMM in the frequency domain, showing consistent pole locations and damping ratios. The study introduces PF analysis in the time domain to link each mode to its spatial influence within the network. While frequency-domain tools such as the FRD describe resonance behavior, PF analysis reveals how nodal heads and pipe flows contribute to each mode, highlighting locations that are more sensitive to excitation. Time-domain simulations show that inputs applied at high-PF locations generate strong transient responses near the system's fundamental frequency, whereas low-PF excitations lead to minimal response. These results confirm the value of PF analysis for identifying regions where modest disturbances may trigger significant pressure surges due to resonance. The New York tunnel case study demonstrates the practicality of the proposed approach for large-scale networks. Combining modal and PF analyses exposes spatial patterns of vulnerability not evident in frequency analysis alone. From an engineering perspective, PF-based indicators can support resilience assessment by identifying areas more susceptible to transient disturbances, such as sudden demand changes, pump failures, or valve operations. Future work focuses on integrating PF-based monitoring into real-time applications and developing targeted control strategies for transient suppression.

Appendix A: RWCM Eigen Analysis

In the RWCM, fewer state variables are used compared to the EWCM. Specifically, in the RWCM, the state variables are limited to independent flow variables. This reduction in the number of state variables presents challenges when deriving the Jacobian matrix from the momentum equation, as the interconnection between the variables complicates the process. The RWCM assumes an infinite wave speed, a key assumption that simplifies the fundamental governing equations, which are expressed as follows (Shimada, 1992):

$$L \frac{dq}{dt} = -R \text{diag}\{|q|\} q + A_I^T h_I + A_R^T h_R \quad (\text{A1})$$

$$A_I q = -Q_I \quad (\text{A2})$$

The derivation starts by differentiating Equation A2 and substituting Equation A1, leading to an expression for the internal heads h_I :

$$h_I = K^{-1} \left(A_I L^{-1} [R \text{diag}\{|q|\} q - A_R^T h_R] - \frac{Q_I(t)}{dt} \right) \quad (\text{A3})$$

Here, $K = A_I L^{-1} A_I^T$ and, when h_I is substituted back into Equation A1, the following expression is derived:

$$\frac{\partial q}{\partial t} = L^{-1} \{ -A_I^T K^{-1} A_I L^{-1} - I \} R \text{diag}\{|q|\} q - L^{-1} A_I^T K^{-1} \frac{Q_I(t)}{dt} + L^{-1} \{ A_R^T h_R - A_I^T K^{-1} A_I L^{-1} A_R^T h_R \} \quad (\text{A4})$$

Defining $W = L^{-1} (I - A_I^T K^{-1} A_I L^{-1})$, and incorporating the permutation matrix P , which organizes the state variables into independent and dependent flows, leads to the refined equation:

$$P \frac{dq}{dt} = -PWP^{-1} P [R \text{diag}\{|q|\} q - A_R^T h_R] + PL^{-1} A_I^T K^{-1} \frac{dQ_I}{dt} \quad (\text{A5})$$

The first n_i rows of this equation set form a set of ODEs for RWCM, allowing the determination of q_i , the independent flow rates. Equation A6 simplifies the matrix expression, while Equation A7 provides the dynamics of q_i :

$$PWP^{-1} = \begin{pmatrix} W_i \\ W_d \end{pmatrix} \quad (\text{A6})$$

$$\frac{dq_i}{dt} = -W_i P [R \text{diag}\{|q|\} q - A_R^T h_R] + PL^{-1} A_I^T R^{-1} \frac{dQ_I}{dt} \quad (\text{A7})$$

Alternatively, the flow vector q is partitioned into q_i (independent flow rates) and q_d (dependent flow rates), where q_d is expressed as a function of q_i , following Shimada (1992):

$$q_d = -A_{Id}^{-1} (A_{Ii} q_i + Q_I) \quad (\text{A8})$$

Two additional matrices, E_{i1}^T and E_{i2}^T , are introduced to decompose W_i as follows:

$$E_{i1}^T = [I_{ni} \ 0_{n_i \times n_D}] \quad (\text{A9})$$

$$E_{i2}^T = [0_{n_D \times n_i} \ I_{n_D}] \quad (\text{A10})$$

Here, n_D represents the number of dependent pipes, and n_i represents the number of independent pipes. By applying Equation A9 and A10 to the first term on the right-hand side of Equation A7 the following relationship is established:

$$W_i P [-R \text{diag}\{|q|\} q] = -W_{i1} R \text{diag}\{q_i\} q_i - W_{i2} R \text{diag}\{q_d\} q_d \quad (\text{A11})$$

$$\begin{aligned} W_{i1} &= W_i E_{i1}^T \\ W_{i2} &= W_i E_{i2}^T \end{aligned} \quad (\text{A12})$$

Equation A8 is substituted into Equation A7, resulting in:

$$W_i P [-R \text{diag}\{|q|\} q] = -W_{i1} R \text{diag}\{q_i\} q_i - W_{i2} R \text{diag}\{q_d\} (-A_{Id}^{-1} (A_{Ii} q_i + Q_I)) \quad (\text{A13})$$

Substituting Equation A13 into Equation A7 and taking the derivative, the Jacobian matrix is derived as follows:

$$J = -2W_{i1} R_i \text{diag}\{|q_i|\} - 2W_{i2} \text{diag}\{R_d\} - A_{Id}^{-1} (A_{Ii} q_i + Q_I) \} (-A_{Id}^{-1} (A_{Ii})) \quad (\text{A14})$$

The eigenvalues of the Jacobian matrix are real and negative, corresponding to the damped modes of the system. These eigenvalues represent the decay rates of the system's dynamic response, showing how quickly transient events are damped in the RWCM.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The MATLAB code developed and used in this study to implement the EWCM and perform modal and PF analyses of WDSs is publicly available at Zenodo (Imani, 2025). The code is released under the MIT license and includes the scripts used to reproduce all simulation results and figures in this paper.

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References

- Abdulrahman, I. (2020). MATLAB-Based programs for power System dynamic analysis. *IEEE Open Access Journal of Power and Energy*, 7, 59–69. <https://doi.org/10.1109/OAJPE.2019.2954205>
- Abed, E. H., Lindsay, D., & Hashlamoun, W. A. (2000). On participation factors for linear systems. *Automatica*, 36(10), 1489–1496. [https://doi.org/10.1016/S0005-1098\(00\)00082-0](https://doi.org/10.1016/S0005-1098(00)00082-0)
- Bohorquez, J., Lambert Martin, F., & Simpson Angus, R. (2020). Identifying head accumulation due to transient wave superposition in pipelines. *Journal of Hydraulic Engineering*, 146(1), 04019044. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001631](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001631)
- Che, T.-C., Duan, H.-F., & Lee, P. J. (2021). Transient wave-based methods for anomaly detection in fluid pipes: A review. *Mechanical Systems and Signal Processing*, 160, 107874. <https://doi.org/10.1016/j.ymssp.2021.107874>
- Dandy, G. C., Simpson, A. R., & Murphy, L. J. (1996). An improved genetic Algorithm for pipe network optimization. *Water Resources Research*, 32(2), 449–458. <https://doi.org/10.1029/95WR02917>

- Dussaud, F. (2015). *An application of modal analysis in electric power systems to study inter-area oscillations* KTHJ. <https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-163404>
- Franklin, G., Powell, J., & Emami-Naeini, A. (2010). *Feedback control of dynamic systems*. Prentice-Hall.
- Garofalo, F., Iannelli, L., & Vasca, F. (2002). Participation factors and their connections to residues and relative gain array. *IFAC Proceedings Volumes*, 35(1), 125–130. <https://doi.org/10.3182/20020721-6-ES-1901.00182>
- Gong, J., Stephens Mark, L., Lambert Martin, F., Zecchin Aaron, C., & Simpson Angus, R. (2018). Pressure surge suppression using a metallic-plastic-metallic pipe configuration. *Journal of Hydraulic Engineering*, 144(6), 04018025. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001468](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001468)
- Gong, Simpson, A., Lambert, M., Zecchin, A., Kim, Y.-i., & Tijsseling, A. (2013). Detection of distributed deterioration in single pipes using transient reflections. *Journal of Pipeline Systems Engineering and Practice*, 4(1), 32–40. [https://doi.org/10.1061/\(ASCE\)PS.1949-1204.0000111](https://doi.org/10.1061/(ASCE)PS.1949-1204.0000111)
- He, J., & Fu, Z.-F. (2001). 1 - Overview of modal analysis. In J. He & Z.-F. Fu (Eds.), *Modal analysis* (pp. 1–11). Butterworth-Heinemann. <https://doi.org/10.1016/B978-075065079-3/50001-2>
- Imani, M. (2025). Morteza2400/ModalAnalysisOfWDSs: V1.0.0. <https://doi.org/10.5281/zenodo.15487412>
- Imani, M., Zecchin, A., Zeng, W., & Lambert, M. F. (2025). Generalization and analysis of elastic water column model for hydraulic transient analysis of water distribution systems. *Journal of Water Resources Planning and Management*, 151(10), 04025054. <https://doi.org/10.1061/JWRMD5.WRENG-6946>
- Imani, M., Zecchin, A., Zeng, W., & Lambert, M. F. (2026). Real-time estimation of states and unknown boundary conditions in pressurized pipeline systems. *Mechanical Systems and Signal Processing*, 242, 113664. <https://doi.org/10.1016/j.ymssp.2025.113664>
- Jung, B. S. (2022). Water distribution System optimization accounting for worst-case transient loadings (pp. 20–29).
- Jung, B. S., Boulos, P. F., & Altman, T. (2011). Optimal transient network design: A multi-objective approach. *Journal American Water Works Association*, 103(4), 118–127. <https://doi.org/10.1002/j.1551-8833.2011.tb11440.x>
- Jung Bong, S., & Karney Bryan, W. (2009). Systematic surge protection for worst-case transient loadings in water distribution systems. *Journal of Hydraulic Engineering*, 135(3), 218–223. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2009\)135:3\(218\)](https://doi.org/10.1061/(ASCE)0733-9429(2009)135:3(218))
- Klein, M., Rogers, G. J., & Kundur, P. (1991). A fundamental study of inter-area oscillations in power systems. *IEEE Transactions on Power Systems*, 6(3), 914–921. <https://doi.org/10.1109/59.119229>
- Lee, P. J., Lambert, M. F., Simpson, A. R., Vitkovský, J. P., & Liggett, J. (2006). Experimental verification of the frequency response method for pipeline leak detection. *Journal of Hydraulic Research*, 44(5), 693–707. <https://doi.org/10.1080/00221686.2006.9521718>
- Lee, P. J., Vitkovský, J. P., Lambert, M. F., Simpson, A. R., & Liggett, J. A. (2005). Frequency Domain analysis for detecting pipeline leaks. *Journal of Hydraulic Engineering*, 131(7), 596–604. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2005\)131:7\(596\)](https://doi.org/10.1061/(ASCE)0733-9429(2005)131:7(596))
- Lee Pedro, J., Vitkovský John, P., Lambert Martin, F., & Simpson Angus, R. (2008). Valve design for extracting response functions from hydraulic systems using Pseudorandom binary signals. *Journal of Hydraulic Engineering*, 134(6), 858–864. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2008\)134:6\(858\)](https://doi.org/10.1061/(ASCE)0733-9429(2008)134:6(858))
- Louati, M., Ghidaoui Mohamed, S., Tekitek Mohamed, M., & Jose Lee, P. (2020). Wave-Leak interaction in a simple pipe System. *Journal of Hydraulic Engineering*, 146(4), 04020013. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001714](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001714)
- Shimada, M. (1992). State-Space analysis and control of slow transients in pipes. *Journal of Hydraulic Engineering*, 118(9), 1287–1304. [https://doi.org/10.1061/\(ASCE\)0733-9429\(1992\)118:9\(1287\)](https://doi.org/10.1061/(ASCE)0733-9429(1992)118:9(1287))
- Souza, O. H., Barbieri, N., & Santos, A. H. M. (1999). Study of hydraulic transients in hydropower plants through simulation of nonlinear model of penstock and hydraulic turbine model. *IEEE Transactions on Power Systems*, 14(4), 1269–1272. <https://doi.org/10.1109/59.801883>
- Ulanicki, B., & Beaujean, P. (2021). Modeling dynamic behavior of water distribution systems for control purposes. *Journal of Water Resources Planning and Management*, 147(8), 04021043. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001403](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001403)
- Wylie, E. B., & Streeter, V. L. (1993). *Fluid transients in systems*. Prentice Hall.
- Xing, L., & Sela, L. (2020). Transient simulations in water distribution networks: TSNet python package. *Advances in Engineering Software*, 149, 102884. <https://doi.org/10.1016/j.advengsoft.2020.102884>
- Zecchin, A., Kim, S., & Ferrante, M. (2018). A modal-based analysis of pipeline networks with applications to time-domain simulation 13th International Conference on Pressure surges.
- Zecchin, A. C. (2010). Laplace-domain analysis of fluid line networks with applications to time-domain simulation and system parameter identification Ph.D. thesis, School of Civil, Environmental and Mining Engineering, Univ. of Adelaide].
- Zecchin, A. C., Simpson, A. R., Lambert, M. F., White, L. B., & Vitkovský, J. P. (2009). Transient modeling of arbitrary pipe networks by a laplace-domain admittance matrix. *Journal of Engineering Mechanics*, 135(6), 538–547. [https://doi.org/10.1061/\(ASCE\)0733-9399\(2009\)135:6\(538\)](https://doi.org/10.1061/(ASCE)0733-9399(2009)135:6(538))
- Zeng, W., Gong, J., Zecchin Aaron, C., Lambert Martin, F., Simpson Angus, R., & Cazzolato Benjamin, S. (2018). Condition assessment of water pipelines using a modified layer-peeling method. *Journal of Hydraulic Engineering*, 144(12), 04018076. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001547](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001547)
- Zeng, W., Zecchin Aaron, C., & Lambert Martin, F. (2022). Elastic water column model for hydraulic transient analysis of pipe networks. *Journal of Hydraulic Engineering*, 148(12), 04022027. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0002028](https://doi.org/10.1061/(ASCE)HY.1943-7900.0002028)